

A Different Image Content-based Retrievals using OCR Techniques

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Abstract- It is very difficult to retrieve image from large no of database which contain some message on it. The OCR techniques are becoming very efficient techniques for external and fast retrievals. The OCR technique search image based on data or text written on image. It search image or message contain in the image OCR the mechanical or electronic conversion of images of typed, handwritten or printed text, whether from scanned document, a scene-photo. For that Tesseract will use. After recognizing text from image by OCR, it will store that message in any file. After that we used Boyer–Moore string search algorithm, for searching string stored in file.

Keyword- *OCR is Optical Character Recognition, Tesseract is a search engine, TBIR- Text Based Image Retrievals, CBIR- Contain Based Image Retrievals.*

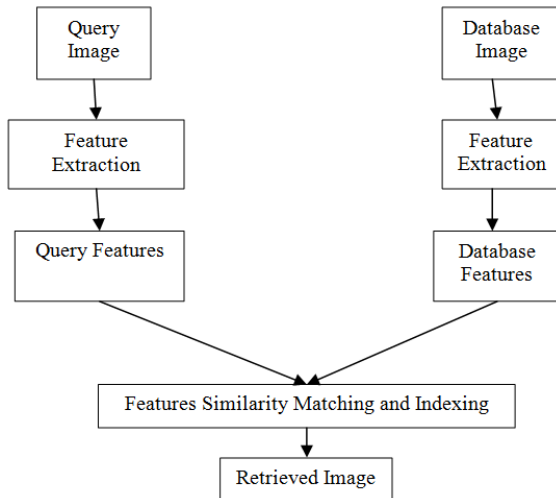
1. INTRODUCTION

The size of the digital image collection has increased rapidly it is very difficult for storing and retrieving digital images which have different application. Therefore an efficient image retrieval system is required. Image retrievals is a process of searching image from large number of data set. The image retrieval system is classified into two types such as Text based image retrieval and Content based image retrieval. The text based image retrieval was introduced in the year 1970s in this method the images are manually tagged by text

descriptors which are used by database management system (DBMS) for performing image retrieval. But the text based retrieval method has a lot of demerits such as loss of information, time consuming, more expensive task, inaccuracy and decreased efficiency. To overcome these problems the content based image retrieval method is proposed. In addition to that the OCR will be used for a full text analysis, to retrieve text from image. OCR is Optical Character Recognition (Optical Character Reader, OCR) is the mechanical or electronic conversion of images of typed, handwritten or printed text, whether from scanned document, a scene-photo. For this we apply Tesseract-OCR, optical character recognition software, to image spam mail filter. Tesseract-OCR is free OCR software and licensed with the Apache 2.0 License. Tesseract-OCR can be specific to a certain language and it specializes Tesseract-OCR to spam words that are used in spam filters such as Bayesian filter. This specialization decreases time and CPU power that it takes to check whether images include spam words.

A. Content Based Image retrieval

Content Based Image retrieval is the process of searching and retrieving images from the huge set



of database based on the visual content such as colour, texture, shape and edge on the basis of user's request by means of query. Search is performed using the actual content like the colour, shapes and texture of the image. "Image retrieval" means searching and retrieving images from large datasets of digital images, the main objective of CBIR is increased retrieval accuracy with the reduced retrieval time. In order to meet the above requirements.

Fig 1: Process flow of Image Retrieval System

B Text Based image retrievals (TBIR)

TBIR is very popular framework. The text based image retrieval was introduced in the year 1970s. TBIR was first annotated the images by text and then used text-based database management systems to perform image retrieval. TBIR is used to manually annotate the image in the database with Annotations keywords or Descriptions. This process is used to describe both image contents and other metadata of the image.

C Optical Character Recognition (OCR)

OCR is Optical Character Recognition (Optical Character Reader, OCR) is the mechanical or electronic conversion of images of typed, handwritten or printed text, whether from

scanned document, a scene-photo. Tesseract-OCR, optical character recognition software, to image spam mail filter. Tesseract-OCR is free OCR software and licensed with the Apache 2.0 License. Tesseract was probably the first OCR engine able to handle white-on-black text so trivially. At this stage, outlines are gathered together, purely by nesting, into *Blobs*.

Blobs are organized into text lines, and the lines and regions are analyzed for fixed pitch or proportional text. Text lines are broken into words differently according to the kind of character spacing. Fixed pitch text is chopped immediately by character cells. Proportional text is broken into words using definite spaces and fuzzy spaces.

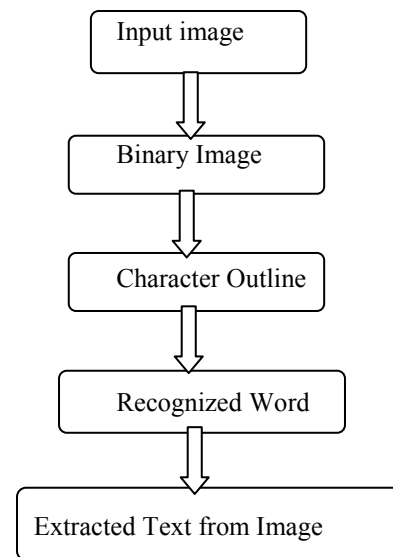


Fig 2: Architecture of Tesseract OCR

2 COMMON FEATURES FOR IMAGE RETRIEVAL

A feature is defined as capturing a certain visual property of an image. In general, image features can be either global or local. The global features describe the visual content of the entire image, whereas local features describe the regions or objects of the image content. The advantage of

global extraction is its high speed for both extracting features and computing similarity. However, global features are often too rigid to represent an image.

A. Colour Features-

The colour has widely been used in IR systems, because of its easy and fast computation. Colour is also an intuitive feature and plays an important role in image matching. Most IR systems use colour space, histogram, moments, colour coherence vector, and dominant colour descriptor represent colour. The colour histogram is one of the most commonly used colour feature representation in image retrieval.

B. Texture Feature

Texture is a property that represents the surface and structure of an image. Texture can be defined as a regular repetition of an element or pattern on a surface. Image textures are complex visual patterns composed of entities or regions with sub-patterns with the characteristics of brightness, colour, shape, size...etc.

C. Shape Features

Shape can generally be defined as the description of an object regardless of its position, orientation, and size. Therefore, shape features should be invariant to translation, rotation, and scale for an effective IR. In the direction of using shape as an image feature, it is necessary to determine object or region boundaries in the image and this is a challenge. Compared with colour and texture features, shape features are usually described after images have been segmented into regions or objects.

D. Spatial Location Features

Spatial location is also significant and is used for region segmentation. Spatial location is described as top/bottom, top left/right and back/front as per the position of an object in an image. If some time

different object contain same colour but spatial information is not similar, so it gives the different position of the image.

2 LITERATURE REVIEW

Several reviews of the literature on image retrieval have been published, from a variety of different viewpoints. K. B. Jayarraman, Salomia Brighitha.J "Survey on Content Based Image Retrieval Technique" [2] various techniques of CBIR such as k-means clustering, k-nearest neighbours Algorithm (KNN), colour structure descriptor (CSD), Text based image retrieval (TBIR) techniques which increase the effectiveness of fast retrieval are discussed and analysed. in this CBIR method has been widely used in various areas to improve the performance of the system and achieve better results in different applications. They prove the efficiency of the image retrieval is improved by means of K-Means clustering algorithm.

Mohammed Alkhawani and Mohammed Elmogy survey on "Text-based, Content-based, and Semantic-based Image Retrievals",[1] It present a survey and discuss the current literature of different types of image retrieval (IR) systems. An overview of the important techniques in image retrieval will be discussed. Finally, some urgent challenges in IR, that have been raised recently, will be presented as well as possible directions for future research. In this they discuss object recognition and detection is one of the most challenging problems in image retrieval.

T. Ojala, M. Pietikainen, and T. Maenpaa, this introduces a theoretically and computationally simple yet efficient multiresolution approach to gray-scale and rotation invariant texture classification based on uniform local binary patterns and nonparametric discrimination of

sample and prototype distributions[4] They developed a generalized gray-scale and rotation invariant operator LBP, which allows for detecting uniform patterns in circular neighborhoods of any quantization of the angular space and at any spatial resolution.

Suraya Abu Bakar, Muhammad Suzuri Hitam, Wan Nural Jawahir Hj Wan Yussof, "Content-Based Image Retrieval using SIFT for Binary and Greyscale Images"[5] it gives an alternative approach for Content Based Image Retrieval (CBIR) using Scale Invariant Feature Transform (SIFT) algorithm for binary and gray scale images. The motivation to use SIFT algorithm for CBIR is due to the fact that SIFT is invariant to scale, rotation and translation as well as partially invariant to affine distortion and illumination changes. Inspired by these facts, they investigate the fundamental properties of SIFT for robust CBIR by using MPEG-7, COIL-20 and ZuBuD image databases. Here it detected key points and its descriptors to match between the query images and images from the database. And their experimental results show that the proposed CBIR using SIFT algorithm producing excellent retrieval result for images with many corners as compared to retrieving image with less corners.

Luke Ledwich, Stefan Williams [6] Reduced SIFT Features for Image Retrieval and Indoor Localization the SIFT feature description of an image is typically large and slow to compute. They present a method to reduce the size, complexity and matching time of SIFT feature sets for use in indoor image retrieval and robot localization. This method takes advantage of the structure of typical indoor environments to reduce the complexity of each SIFT feature and the number of SIFT features required to describe a scene.

Liao, M.W. K. Law, and A. C. S. Chung, [3] they propose the dominant local binary patterns (DLBP) as a texture classification approach. The DLBP approach on one side guarantees to be able to represent the dominant patterns in the texture images. On the other side, it retains the rotation invariant and histogram equalization invariant properties of the conventional LBP approach. It is simple and computationally efficient.

3 MOTIVATIONS

In recent years, collections of digital images are created and increased rapidly. In many areas of academia, commerce, government, medicine, and Internet, a huge amount of information is out there. However, we cannot access or make use of this information unless it is organized to allow efficient browsing, searching, and retrieval. One of the main problems is the difficulty of locating a desired image in a large and varied collection. While it is perfectly feasible to identify a desired image from a small collection simply by browsing, more effective techniques are needed with collections containing thousands of items. Image retrieval attracts interest among researchers in the fields of image processing, multimedia, digital libraries, remote sensing, astronomy, database applications, and other related areas There is need to effective ways retrieve of similar images that are conform to human perception and without human interference. In image annotation some issues poses a challenge: images have to be automatically annotated with meaningful labels.

Eakins proposes a framework for image retrieval classifying image queries into a series of levels, and discussing the extent to which advances in technology are likely to meet users' needs at each level. His conclusion is that automatic CBIR techniques can already address many of users' requirements at level 1, and will be capable of

making a significant contribution at level 2 if current research ideas can be successfully exploited. They are however most unlikely to make any impact at level 3 in the foreseeable future.

Enser reviews methods for providing subject access to pictorial data, developing a four-category framework to classify different approaches. He discusses the strengths and limitations both of conventional methods based on linguistic cues for both indexing and search, and experimental systems using visual cues for one or both of these. His conclusions are that, while there are serious limitations in current text-based techniques for subject access to image data, significant research advances will be needed before visually-based methods are adequate for this task. He also notes, as does Cawkell in an earlier study, that more dialogue between researchers into image analysis and information retrieval is needed.

4 PROBLEM DEFINITIONS

There are many issues of intense research that still unsolved or solved with limited success for example: The optimization of feature processing time and query response time are important for a huge image database. The selection of algorithms, parameters etc. is very important because specific one is not necessary suitable for all application e.g. the segmentation algorithm used for natural images may not be suitable for medical images. An appropriate index structure that allows efficient searching of large image database is still a problem under research. On the other hand, the object recognition and detection is one of the most challenging problems in image retrieval. The addressing of this problem is urgent. In general, the object should be recognized regardless of the illumination changes, changes of size, rotation, background clutter, viewpoint change, and occlusion. However, this cannot be obtained in

most CBIR because it does not have adequate ability to capture important properties of object. In support of better object recognition, the interest point's detectors were introduced to represent the local features of images in image retrieval systems. Binary histogram had also been proposed that can be used for image classification and retrieval without image segmentation which provides good retrieval performances, but it only works on gray level images.

5 OBJECTIVES

- To Extract text from image for efficient searching, and Remove language restriction while content based searching.
- To Searching for image from large number of images that contain same object or same scene with different viewpoints.
- To Retrieves similar images from large image database based on image colour.
- To Retrieves images according to texture of images.
- To retrieves images according to similar shapes from image.

6 PROPOSED WORK

Many feature like translation and rotation cannot be retrieving by CBIR. So Scale Invariant Features transform or widely known as SIFT is one of the techniques that have been successfully used for local interest point detector and its descriptors. Scale-invariant feature transform (or SIFT) is an algorithm in computer vision to detect and describe local features in images. Local image features are specific image properties of local image region such as edges, corners, lines and curves. SIFT is not just scale invariant but also invariant to rotation and translation.

At the same time we give feature for image retrieval on character based. It search image on message contain in the image. For that Optical Character Recognition is used. OCR the mechanical or electronic conversion of images of typed, handwritten or printed text, whether from scanned document, a scene-photo. For that Tesseract will use. After recognizing text from image by OCR, it will store that message in any file. After that we used Boyer–Moore string search algorithm, for searching string stored in file. The Boyer–Moore string search algorithm is an efficient string searching algorithm that is the standard benchmark for practical string search literature. It was developed by Robert S. Boyer and J. Strother Moore in 1977. The algorithm preprocesses the string being searched for (the pattern), but not the string being searched in (the text). The algorithm runs faster as the pattern length increases.

For this software requirements are java and jdk 1.7. Ide, netbean 6.9 and operating system is windows x. From the following architecture shows the how OCR work with other feature for image retrievals.

From the diagram it shows that, image can be retrieve with different feature of text, colour and texture along with it can be retrieve with the text contain in that image. This architecture shows with the help of OCR, it can be recognized characters. The Fig 3: shows the process flow of the image retrieval system and is described as follows:

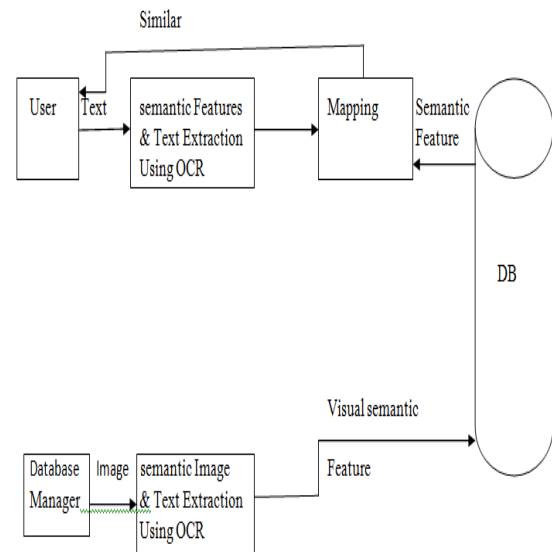


Fig 3: Typical Semantic-based & Text-based Image Retrieval

- **Collection of Database:** The database contains the collection of images which are stored in any one of the formats .jpg, .bmp, .tiff.
- **Query:** The user provides the sample image or the text as a query for the system.
- **Feature extraction:** There are various kinds of visual features to represent an image such as colour, texture, size and spatial relationship. Single feature can represent the part of the image property. Hence the combination of all the features of an image is used for effective image retrieval.
- **Similarity mapping:**
- **Text Extraction using OCR:** In this it retrieves the text from the image and stored in data base.
- **Retrieval:** The system retrieves the images based on the sequence of ranked images, or given text from the database.

7. CONCLUSION

In this paper we have discussed various method and techniques of image retrievals with different parameter. CBIR method has been widely used in various areas to improve the performance of the

system and achieve better results in different applications. More key points of SIFT features will be detected for image that have corners and edges which make the retrieval results better than image that has lesser corners. Our experiment show that OCR is search image which contain text message on it, it is alternative way of image retrieval by using OCR. Text retrieval is the basis of image retrieval, Many techniques come from this domain. Text and image feature of retrievals combined, have biggest chances for success.

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References

- [1] Mohammed Alkhawlan, Mohammed Elmogy Jayarraman, "International Journal of Advanced Research in Computer Science and Software Engineering", March-2016, pp. 587-591
- [2] K. B. Jayarraman, Salomia Brighitha, "International Journal of Computer and Information Technology" (ISSN: 2279 – 0764) Volume 04 – Issue 01, January 2015
- [3] S. Liao, M.W. K. Law, and A. C. S. Chung, "Dominant local binary patterns for texture classification," *IEEE Trans. Image Process.*, vol. 18, no. 5, pp. 1107–1118, May 2009.
- [4] T. Ojala, M. Pietikainen, and T. Maenpaa, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," *IEEE Trans Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 971–987, Jul. 2002.
- [5] Suraya Abu Bakar, Muhammad Suzuri Hitam, Wan Nural Jawahir Hj Wan Yussof "Content-Based Image Retrieval using SIFT for Binary and Greyscale Images" 978-1-4799-0269-9/13/\$31.00 ©2013 IEEE
- [6] Luke Ledwich, Stefan Williams, "Reduced SIFT Features For Image Retrieval and Indoor Localisation", *International Journal of Computer Vision*, 60, 2 (2004), pp. 91-110.
- [7] Eakins, J. P. and Graham, M. E. 2000. "Content based image retrieval. Technical Report JTAP-039. JISC Technology Application Program, Newcastle upon Tyne".
- [8] Enser, P.G.B. and McGregor, C.G. 1992. "Analysis of visual information retrieval queries. Personal Communication".
- [9] S.Mangijao Singh, K. Hemachandran, "Content- Based Image Retrieval using Color Moment and Gabor Texture Feature", *IJCSI International Journal of Computer Science Issues*, Vol.9, Issue 5, No 1, September 2012 ISSN:1694-8014
- [10] M. Singha and K. Hemachandran, "Content based image retrieval using color and texture," *Signal & Image Processing: An International Journal (SIPIJ)*, vol. 3, 2012, pp. 39–57.
- [11] M. Pietikainen, T. Ojala, T. Scruggs, K. W. Bowyer, C. Jin, K. Hoffman, J. Marques, M. Jacsik, and W. Worek, "Rotational invariant texture classification using feature distributions," *Pattern Recogn.*, vol. 33, no. 1, pp. 43–52, Jan. 2000.
- [12] Vivek Jain, Neha Sahu, "Content Based Image Retrieval, *International Journal of Engineering Research and Application*, Vol 8, Issue 4, July- August 2013, pp.1166-1169.